WORKED SOLUTION

STEP II 2003, Question 12

Probability and Statistics • Branching Processes

STEP II 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q12

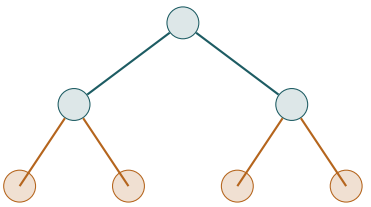
This question has a full guided treatment in *Beyond STEP*, Section 12.15, *Extinction or Explosion*: https://beyondstep.keithmcnulty.org/ch12/branching.html

The key idea

This is a branching process: each generation is built from the previous one, with every particle splitting independently. The one fact that does all the work is that if m particles each produce a $\mathrm{B}(2, p)$ number of offspring, independently, the total is $\mathrm{B}(2m, p)$. So we condition on the size of the previous generation, both for the probabilities and for the expectation.

Solution

We assume, as the model intends, that different particles disintegrate independently of each other and of the past. Write $X_0 = 1$ for the single particle alive during $0 < t < T$. At $t = nT$ each of the X_{n-1} particles alive is replaced by its own offspring, so X_n is the total number of offspring of generation $n - 1$.



$X_0 = 1$

$X_1 = 2$

$X_2 \leq 4$

The picture shows the case $X_1 = 2$: the two particles then have up to four offspring between them, and each of the four possible offspring is born with probability p, independently.

The joint probability

Since $X_1 \sim \mathrm{B}(2, p)$, we have $\mathrm{P}(X_1 = 2) = p^2$. Given $X_1 = 2$, each of the two particles produces a $\mathrm{B}(2, p)$ number of offspring, independently, so X_2 is the number of successes in 4 independent trials with success probability p; that is, $X_2 \mid X_1 = 2 \sim \mathrm{B}(4, p)$. Hence

$$\mathrm{P}(X_1 = 2 \text{ and } X_2 = 2) = \mathrm{P}(X_1 = 2)\,\mathrm{P}(X_2 = 2 \mid X_1 = 2) = p^2 \cdot \binom{4}{2} p^2 q^2 = 6p^4 q^2,$$

as required.

The conditional probability

To use the definition of conditional probability we need $\mathrm{P}(X_2 = 2)$, which we find by conditioning on X_1:

- if $X_1 = 0$ the population has died out, so $X_2 = 0$;
- if $X_1 = 1$ (probability $2pq$), then $X_2 \sim \mathrm{B}(2, p)$ and $\mathrm{P}(X_2 = 2 \mid X_1 = 1) = p^2$;
- if $X_1 = 2$, the contribution is the $6p^4q^2$ just found.

So

$$\mathrm{P}(X_2 = 2) = 2pq \cdot p^2 + 6p^4q^2 = 2p^3q\,(1 + 3pq),$$

and therefore, for $0 < p < 1$,

$$\mathrm{P}(X_1 = 2 \mid X_2 = 2) = \frac{6p^4q^2}{2p^3q(1 + 3pq)} = \frac{3pq}{1 + 3pq}.$$

Setting this equal to $\frac{9}{25}$ gives $75pq = 9 + 27pq$, so $48pq = 9$ and $pq = \frac{3}{16}$. With $q = 1 - p$ this is

$$p^2 - p + \tfrac{3}{16} = 0 \quad \Longleftrightarrow \quad \left(p - \tfrac{1}{4}\right)\left(p - \tfrac{3}{4}\right) = 0,$$

so $p = \frac{1}{4}$ or $p = \frac{3}{4}$. (The cases $p = 0$ and $p = 1$ are excluded: for $p = 0$ the event $X_2 = 2$ is impossible, and for $p = 1$ the conditional probability is 0.)

Ask yourself Why do both values of p give the same answer?

The conditional probability depends on p only through $pq = p(1 - p)$, which is unchanged when p is replaced by $1 - p$. So the roots must come in a pair $p, 1 - p$, here $\frac{1}{4}$ and $\frac{3}{4}$.

The expectation

Suppose $X_{n-1} = m$. Each of these m particles produces on average $\mathrm{E}(X) = 2p$ offspring, so the expected size of the next generation is $2pm$:

$$\mathrm{E}(X_n \mid X_{n-1} = m) = 2pm.$$

Averaging over the possible values of m (the law of total expectation),

$$\mathrm{E}(X_n) = \sum_m \mathrm{P}(X_{n-1} = m)\, 2pm = 2p\,\mathrm{E}(X_{n-1}).$$

In words: every particle alive in one interval contributes $2p$ particles, on average, to the next. Since $\mathrm{E}(X_1) = 2p$ (or $\mathrm{E}(X_0) = 1$), the recurrence gives a geometric sequence:

$$\mathrm{E}(X_n) = (2p)^n.$$

The limits

For $p = \frac{1}{4}$ we get $\mathrm{E}(X_n) = \left(\frac{1}{2}\right)^n \to 0$ as $n \to \infty$, since $0 < \frac{1}{2} < 1$.

For $p = \frac{3}{4}$ we get $\mathrm{E}(X_n) = \left(\frac{3}{2}\right)^n \to +\infty$, since $\frac{3}{2} > 1$.

So the two values of p that look identical in the conditional probability lead to completely different long-term behaviour: on average the population dies away when $p = \frac{1}{4}$ and explodes when $p = \frac{3}{4}$.

Checks

- With $p = \frac{3}{4}$: $6p^4q^2 = \frac{243}{2048} \approx 0.1187$, and a simulation of 10^6 two-generation populations gave 0.1186.
- At $p = \frac{1}{4}$, $pq = \frac{3}{16}$, so $3pq = \frac{9}{16}$ and $\dfrac{9/16}{25/16} = \frac{9}{25}$, as it should be.
- Computing the exact distributions of $X_1, \ldots, X_5$ (each generation $\mathrm{B}(2m, p)$ given the last) for several p gives means exactly $(2p)^n$.

Common slips

- Forgetting the route $X_1 = 1$, $X_2 = 2$ when finding $\mathrm{P}(X_2 = 2)$; the route through $X_1 = 0$ correctly contributes nothing.
- Taking $X_2 \mid X_1 = 2$ to be $2 \times \mathrm{B}(2, p)$ (twice one particle's offspring) rather than a sum of two independent copies, which is $\mathrm{B}(4, p)$.

Beyond STEP

This is a Galton–Watson branching process, first studied to explain why surnames die out. The mean number of offspring $\mu = 2p$ decides everything: if $\mu \leq 1$ the population dies out with probability 1, while if $\mu > 1$ it survives for ever with positive probability. Even for $p = \frac{3}{4}$, the population still dies out with probability $\frac{1}{9}$, the smallest root of $s = (q + ps)^2$, which comes from the probability generating function. Grimmett and Welsh, *Probability: An Introduction*, treat branching processes with generating functions in a way that follows on directly from this question.

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